

Assignment 2

Forces

Communication

10 / 10

Good work!

INSTRUCTIONS

Answer all questions fully and ensure the following:

- Show all your work/steps including all calculations
- Display proper notation for vectors (i.e. include arrows for vector variables)

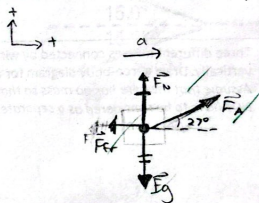
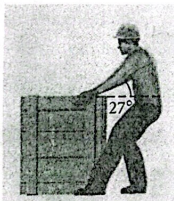
BACKGROUND

Free body diagrams (FBD's) are simplified representations of an object (the body) in a problem, and includes force vectors acting on the object. This body is free because the diagram will show it without its surroundings (i.e. the body is 'free' of its environment). This eliminates unnecessary information which might be given in a problem.

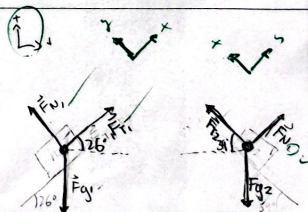
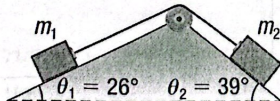
QUESTIONS (COMMUNICATION)

Draw a free-body diagram for the situation shown.

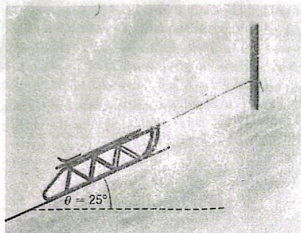
1. A worker pulls a crate at an angle of 27° as shown. Draw a free-body diagram for the crate. Assume a friction between the crate and the floor.



2. Two masses, connected by a massless string, hang over a pulley that connects two inclines. Draw a free-body diagram for each mass. Assume that the string and the pulley have no mass so that they do NOT have to be considered as a separate object. Assume that friction is negligible.



3. A sled is on a hill that is inclined 25° to the horizontal as shown. The hill is very icy (negligible friction), and the sled is held at rest by a rope attached to a post. The rope is parallel to the hill as shown. Draw a free-body diagram for the sled.



4. Three different masses connected by wires and hung vertically. Draw a free-body diagram for only mass B. Assume that the wire has no mass so that they do NOT have to be considered as a separate object.

